



Modelling Financial Markets

Seminar 6: Multivariate Volatility and Dependence — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ US: United States
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Compute tail dependence and copula parameters step by step
- ▣ Estimate a DCC model, test it against constant correlation and read its persistence
- ▣ Fit and test copulas on pseudo-observations with a parametric bootstrap
- ▣ Separate contagion from the volatility bias (Forbes–Rigobon)

- ▣ Background: Lecture 6; textbooks [Franke, Härdle & Hafner \(2019\)](#) and [Nelsen \(2006\)](#)

Structure

- ▣ **Part A:** five pen-and-paper exercises (A1–A5)
- ▣ **Part B:** five computational exercises with inference (B1–B5)
- ▣ **Part C:** one open question, a seed for the team project (C1), and a critique of an AI answer (C2)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- ▣ Report every estimate with its uncertainty: standard error, bootstrap interval or p -value
- ▣ Ask each time whether the standard method can fail here: time zones, boundary parameters, volatility changes, many observations

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Gaussian versus t tail dependence, step by step), A2 (Kendall's tau and copula parameters), A3 (Forbes–Rigobon adjustment, step by step)
- Part B: B1 (DCC on SPY and TLT), B2 (Gaussian versus t copula, S&P 500 and Euro Stoxx 50)

Proposed: you solve them

- Part A: A4 (model: A3), A5 (model: B1)
- Part B: B3 (model: B2, A2), B4 (model: A3), B5 (model: B1)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: A1, A2)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter6_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: SPY, TLT, S&P 500, Euro Stoxx 50, BET, Bitcoin; daily closes up to 18 September 2026
- Joint analyses: prices are first joined on common trading days, then returns are computed
- Weekly returns: last common close of each week (Friday); 2-day returns: $\ln P_t - \ln P_{t-2}$

Formulas Used Today

- DCC: $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$, $R_t = \text{diag}(Q_t)^{-1/2}Q_t\text{diag}(Q_t)^{-1/2}$ Engle (2002)
- Forbes–Rigobon: $\delta = \sigma_{x,c}^2/\sigma_{x,calm}^2 - 1$, $\rho^* = \rho_c/\sqrt{1 + \delta(1 - \rho_c^2)}$ Forbes & Rigobon (2002)
- Pseudo-observations: $\hat{u}_{it} = \text{rank}(x_{it})/(T + 1)$ Genest, Ghoudi & Rivest (1995)
- Tail dependence: $\lambda_L = \lim_{q \downarrow 0} C(q, q)/q$; t copula: $\lambda = 2t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)})$ Demarta & McNeil (2005)
- Kendall's tau: Gaussian/t $\tau = \frac{2}{\pi} \arcsin \rho$; Clayton $\theta/(\theta + 2)$; Gumbel $1 - 1/\theta$
- Goodness of fit: Cramér–von Mises distance of the Rosenblatt pairs $(u, h(v | u))$ from independence, parametric bootstrap Genest, Rémillard & Beaudoin (2009)

A1: Gaussian versus t Tail Dependence, Step by Step [Solved]

Task

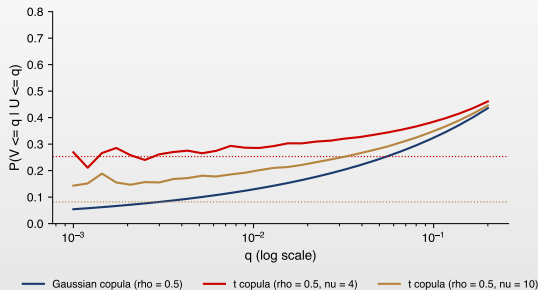
- Two copulas with the same $\rho = 0.5$: Gaussian and Student t with $\nu \in \{3, 4, 10, 30\}$
- (a) Show that the Gaussian copula has $\lambda = 0$ for $|\rho| < 1$ (sketch)
- (b) Compute λ of the t copula for each ν
- (c) At finite levels, compare $P(V \leq q \mid U \leq q)$ for $q = 5\%$ and 0.1%

Solution

- (a) With $x = \Phi^{-1}(q) \rightarrow -\infty$:

$$P(V \leq q \mid U = q) = \Phi\left(x \sqrt{(1 - \rho)/(1 + \rho)}\right) \rightarrow 0, \text{ hence } \lambda = 0$$
- (b) $\sqrt{(\nu + 1) \cdot 0.5/1.5}$: 1.155, 1.291, 1.915, 3.215
- $\lambda = 2t_{\nu+1}(-\cdot)$: 0.313 ($\nu = 3$), 0.253 ($\nu = 4$), 0.082 ($\nu = 10$), 0.003 ($\nu = 30$)
- (c) Gaussian: 0.244 at 5%, 0.054 at 0.1%; t with $\nu = 4$: 0.338 and 0.259
- At 5% the two copulas look alike; deep in the tail the Gaussian probability collapses, the t one does not

A1: The Picture [Solved]



- Exact $P(V \leq q | U \leq q)$ from the bivariate Normal and t distribution functions; dotted lines: the limits λ
- A Gaussian risk model calibrated at the 5% level misses the joint crashes at the 0.1% level

A2: Kendall's Tau and Copula Parameters [Solved]

Task

- S&P 500 and Euro Stoxx 50, weekly returns 2000–2026 ($n = 1\,393$)
- Sample: Kendall's $\hat{\tau} = 0.567$, Spearman $\hat{\rho}_S = 0.749$, Pearson $\hat{\rho} = 0.783$
- (a) Invert τ for the Gaussian, Clayton, Gumbel and Frank copulas
- (b) Implied tail dependence of each
- (c) Check: Spearman implied by the Gaussian copula versus the sample value

Solution

- Gaussian: $\rho = \sin(\pi\hat{\tau}/2) = 0.777$
- Clayton: $\theta = 2\hat{\tau}/(1 - \hat{\tau}) = 2.617$, $\lambda_L = 2^{-1/\theta} = 0.767$
- Gumbel: $\theta = 1/(1 - \hat{\tau}) = 2.308$, $\lambda_U = 2 - 2^{1/\theta} = 0.650$
- Frank: solve $1 - \frac{4}{\theta}[1 - D_1(\theta)] = \hat{\tau}$ numerically: $\theta = 7.104$, no tail dependence
- (c) $\frac{6}{\pi} \arcsin(0.777/2) = 0.762$ versus 0.749: close, so τ alone cannot choose the family

A3: The Forbes–Rigobon Adjustment, Step by Step [Solved]

Task

- Calm period: correlation 0.40; crisis: correlation 0.60
- The variance of the source market is 4 times higher in the crisis
- (a) Compute δ and the adjusted crisis correlation ρ^*
- (b) Which correlation would we measure in the crisis if the true correlation stayed 0.40?
- (c) Contagion or interdependence?

Solution

- $\delta = 4 - 1 = 3$
- $\rho^* = 0.60 / \sqrt{1 + 3(1 - 0.36)} = 0.60 / \sqrt{2.92} = 0.351$
- $\rho_c = 0.40 \sqrt{(1 + 3) / (1 + 3 \cdot 0.16)} = 0.40 \sqrt{4 / 1.48} = 0.658$
- (c) $\rho^* = 0.351 < 0.40$: the whole increase is explained by volatility, i.e. interdependence
- Volatility alone would lift 0.40 to 0.658, more than the observed 0.60

A4: Deriving the Forbes–Rigobon Bias [Proposed]

Solution discussed in class

Task

- Model $y_t = \alpha + \beta x_t + \varepsilon_t$, $\text{Cov}(x_t, \varepsilon_t) = 0$, constant β and σ_ε^2
- (a) Write ρ^2 as a function of $\beta^2 \sigma_x^2 / \sigma_\varepsilon^2$
- (b) Let $\sigma_{x,c}^2 = (1 + \delta) \sigma_x^2$; derive $\rho_c = \rho \sqrt{(1 + \delta) / (1 + \delta \rho^2)}$
- (c) Invert to obtain ρ^* ; (d) which assumption fails if a global shock raises both σ_x and σ_ε ?
- Model: A3

A5: DCC Algebra [Proposed]

Solution discussed in class

Task

- Take the DCC recursion with $a, b \geq 0$, $a + b < 1$ and $\bar{Q}$ positive definite
- (a) Show that Q_t is positive definite for every t
- (b) If $\mathbb{E}[\varepsilon_t \varepsilon_t^\top] = \bar{Q}$, show that $\mathbb{E}[Q_t] = \bar{Q}$ in the stationary case
- (c) With the B1 estimates $a = 0.0529$, $b = 0.9326$: half-life of a correlation shock; what happens if $a + b = 1$?
- Model: B1

B1: DCC for SPY and TLT [Solved]

Task

- ▣ Daily SPY and TLT, prices joined first, 2002–2026
- ▣ (a) GARCH(1,1) for each series; standardised residuals ε_t
- ▣ (b) DCC(1,1) by maximum likelihood; standard errors; likelihood ratio against CCC
- ▣ (c) Re-estimate on 2002–2019 and 2020–2026
- ▣ (d) Test whether the correlation of ε_t differs before and after January 2022 (Fisher z)

Interpretation

- ▣ Is the persistence $a + b$ stable across subsamples?
- ▣ Why is the χ^2 distribution only indicative for the likelihood ratio?
- ▣ What does a change in $\bar{Q}$ mean for the DCC model on the full sample?

B1: Results [Solved]

	a	b	$a + b$	$\bar{Q}_{12}$	T
2002–2026	0.0529 (0.0060)	0.9326 (0.0084)	0.9856	-0.248	6 023
2002–2019	0.0569	0.9194	0.9764	-0.350	4 342
2020–2026	0.0350	0.9514	0.9863	0.002	1 681

- Likelihood ratio against CCC: 526.4; even against a conservative χ^2_2 reference the rejection is overwhelming
- χ^2 is only indicative: under $a = 0$ the parameter b is not identified (Davies problem)
- Persistence is similar ($a + b \approx 0.98$), but the target moves from -0.35 to 0.00
- Correlation of ε_t : -0.343 before 2022, 0.124 after; Fisher $z = 14.8$: a structural break in $\bar{Q}$
- A full-sample DCC pulls the post-2022 correlation towards a target that no longer applies: estimate on the recent regime or allow $\bar{Q}$ to shift

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50? [Solved]

Task

- Weekly returns 2000–2026 (as in A2); pseudo-observations
- (a) Fit Gaussian and t copulas by canonical maximum likelihood; AIC
- (b) Likelihood ratio for $H_0 : \nu = \infty$; p -value by parametric bootstrap (100 replications)
- (c) Rosenblatt goodness-of-fit test for both
- (d) 90% parametric-bootstrap interval for ν and λ (200 replications)

Interpretation

- Why not use the χ^2_1 distribution for the likelihood ratio?
- Does passing the goodness-of-fit test prove the t copula is right?
- How does $\hat{\lambda}$ compare with the empirical $\hat{C}(q, q)/q$ at $q = 5\%$?

B2: Results [Solved]

	$\hat{\rho}$	$\hat{\nu}$	ℓ	AIC	fit p
Gaussian	0.773	–	630.4	-1258.7	0.005
Student t	0.778	3.98	680.4	-1356.9	0.911

- ▣ Likelihood ratio 100.2, bootstrap $p = 0.010$ (the smallest possible with 100 replications): Gaussian rejected
- ▣ $\nu = \infty$ lies on the boundary of the parameter space, so the χ_1^2 reference is wrong (a $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ mixture is the textbook fix)
- ▣ Fit test: Gaussian rejected ($p = 0.005$), t not rejected ($p = 0.911$); not rejecting is not proof, only absence of evidence against
- ▣ $\hat{\nu} = 3.98$ [2.98, 5.09]; $\hat{\lambda} = 0.466$ [0.415, 0.514]
- ▣ Empirical at $q = 5\%$: lower 0.517, upper 0.488; consistent with $\hat{\lambda}$, with a hint of asymmetry that the t copula ignores

B3: Clayton or Gumbel for Joint Losses of BET and Euro Stoxx 50? [Proposed]

Solution discussed in class

Task

- ▣ Daily BET and Euro Stoxx 50 from 2010 (almost synchronous closes); GARCH(1,1) filter
- ▣ Work with **losses** $L = -\varepsilon$; pseudo-observations of the losses
- ▣ (a) Fit Clayton and Gumbel (and, for reference, Gaussian and t); AIC and tail dependence
- ▣ (b) Rosenblatt goodness-of-fit test with parametric bootstrap for each
- ▣ (c) Empirical $\hat{\lambda}_U(q)$ and $\hat{\lambda}_L(q)$ of the losses at $q = 5\%$ and 10%
- ▣ Model: B2 (fit and test), A2 (parameters and tails)

Interpretation

- ▣ Which corner of the loss copula holds the joint crashes?
- ▣ Why can Clayton on losses not capture joint crashes?
- ▣ What would a rotated (survival) Clayton on losses represent?

B4: Forbes–Rigobon for 2008 and 2020 [Proposed]

Solution discussed in class

Task

- Source market: S&P 500; targets: Euro Stoxx 50 and BET; 2-day returns on common days
- Crises: 15 September 2008 – 31 March 2009 and 20 February – 30 April 2020; calm: the year before each
- (a) ρ_{calm} , ρ_C , δ , ρ^* and one-sided Fisher z tests (effective $n = \text{half the 2-day returns}$)
- (b) Sensitivity: shorten the crisis windows to end-2008 and end-March 2020
- Model: A3

Interpretation

- Where does the raw test find contagion, and does it survive the adjustment?
- Why is δ so much larger in 2020?
- Which assumption of the adjustment is least credible for a pandemic?

B5: Bitcoin and the S&P 500: DCC with Bootstrap Bands [Proposed]

Solution discussed in class

Task

- Bitcoin and S&P 500 on common trading days since September 2014
- (a) GARCH(1,1) + DCC(1,1); average correlation before and after 2020
- (b) Residual bootstrap: decorrelate $\eta_t = L_t^{-1}\varepsilon_t$ (L_t Cholesky factor of R_t), resample η_t , simulate a DCC with $(\hat{a}, \hat{b})$, re-estimate; 200 replications
- (c) 90% band for ρ_t on the original data with the bootstrap parameters
- Model: B1

Interpretation

- Is the post-2020 correlation significantly positive on most days?
- Why does a simple block bootstrap of ε_t bias $\hat{b}$ downwards?
- Which uncertainty does the band ignore?

C1: Did Romanian and Euro-Area Equities Become More Integrated after 2020? [Proposed]

Solution discussed in class

The question

- BET versus Euro Stoxx 50, 2010–2019 against 2020–2026
- 2020 brought COVID-19 and the FTSE Russell upgrade of Romania to Secondary Emerging (FTSE Russell (2019))
- Is a higher correlation after 2020 integration, or only the crises inside the sample?

A good answer includes

- Weekly returns (why?) and a stationary block bootstrap Politis & Romano (1994) for the change in correlation
- A Forbes–Rigobon adjustment for the change in euro-area volatility
- Rank correlation, tail dependence and the BET beta on Euro Stoxx 50
- A robustness check without the February–April 2020 weeks, and a DCC view

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “e holds the GARCH-standardised residuals of JPM and BAC (two US bank stocks). Calibrate a t copula with $\nu = 4$ from Kendall's τ and give me its tail dependence.”

The assistant's answer

```
import numpy as np
from scipy import stats
tau = stats.kendalltau(e['JPM'], e['BAC'])[0]
rho = 2 * np.sin(np.pi * tau / 6)
u = e.rank() / len(e)
z = stats.t.ppf(u, df=4)
nu = 4
x = np.sqrt((nu + 1) * (1 - rho) / (1 + rho))
lam = 2 * stats.t.cdf(-x, df=nu)

# Kendall's tau
# copula correlation
# pseudo-observations
# input of the t-copula likelihood

# tail dependence
```

- “With $\hat{\tau} = 0.50$: $\rho = 0.518$ and $\lambda_L = \lambda_U = 0.276$.”

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs, but the answer contains **three** mistakes typical for copula work
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ Name the line and say why it is wrong
 - ▶ Say how you would detect it: a check by hand, a formula from A1 or A2, a small simulation
 - ▶ Write the corrected line(s) and the corrected numbers
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: what is `u` for the largest residual, and what does `stats.t.ppf` return there?

Common Pitfalls

- ▣ Computing returns before joining series: Monday returns that span a weekend for one asset and not the other
- ▣ Using daily correlations across time zones without 2-day or weekly returns
- ▣ Using χ^2 critical values for tests on the boundary ($a = 0$ in DCC, $\nu = \infty$ in the t copula)
- ▣ Fitting copulas to raw returns instead of GARCH-filtered pseudo-observations
- ▣ Treating a non-rejected copula as the true one, or a rejected one as useless: with thousands of observations every static model is rejected
- ▣ Block-bootstrapping a highly persistent process with short blocks: biased persistence and misleading bands

Take-Home Assignment (Team Project)

- ▣ Choose two markets not covered in class (e.g. WIG20 and BUX, gold and Bitcoin, two bank stocks)
- ▣ Estimate EWMA, rolling and DCC correlations; test DCC against CCC; show the correlation in two crises with the Forbes–Rigobon adjustment
- ▣ Fit five copulas to GARCH-filtered pseudo-observations; report AIC, bootstrap fit tests and tail dependence with intervals
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Measuring Statistical Risk
- ☐  Multivariate Statistical Analysis

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_06

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